# Thomas Verelst

PhD Researcher in Computer Vision - ESAT-PSI, KU Leuven

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Enabling deep learning and computer vision on edge devices.
Strong coding skills, endless curiosity and a touch of entrepreneurship.

Research on model compression, conditional computing and custom CUDA layers for convolutional neural networks.

## EDUCATION

### PhD candidate in Engineering Science

Computer vision and deep learning

Oct 2018 - Present   KU Leuven, Belgium

- PhD promotor: Prof. Tinne Tuytelaars
- Efficient convolutional neural networks (CNN), model compression and dynamic neural networks.
- Implementation of computer vision algorithms in collaboration with industry partners
- Teaching of 'Project & Design' bachelor course and guidance of master students

### Master of Science: Electrical Engineering

Embedded systems and multimedia

Sept 2016 – June 2018   KU Leuven, Belgium

- Magna Cum Laude (81.28%)
- with one-year exchange to EPFL Lausanne, Switzerland

### Bachelor of Science

Electrical engineering - computer science

Sept 2013 – June 2016   KU Leuven, Belgium

- Magna Cum Laude (82.67%)

### Secondary school

Latin-Mathematics

Sint-Jan Berchmanscollege Westmalle, Belgium

## LEAD AUTHOR PUBLICATIONS

### SegBlocks: Towards Block-Based Adaptive Resolution Networks for Fast Segmentation

- Thomas Verelst and Tinne Tuytelaars, ECCV Embedded Vision Workshop 2020 (Best Short Paper Award)
- Reduces computational cost and memory consumption of deep neural networks for segmentation by adapting the processing resolution of image regions.

### Dynamic Convolutions: Exploiting Spatial Sparsity for Faster Inference

- Thomas Verelst and Tinne Tuytelaars, CVPR 2020
- Reduces computational cost of CNNs by selectively applying convolutional operations using end-to-end trained attention.
- https://arxiv.org/abs/1912.03203

### Generating Superpixels using Deep Representations

- Thomas Verelst, Matthew Blaschko and Maxim Berman
- Master thesis work (grade 17.9/20). Extended abstract in DeepVision workshop, CVPR 2018.
- https://arxiv.org/abs/1903.04586

## INTERNSHIPS AND SUMMER SCHOOLS

### ICVSS 2019 summer school

Focusing on computer vision and deep learning

1 week, July 2019   Sicily, Italy

### Internship (master) at Nokia Networks

6 weeks, Summer 2017   Antwerp, Belgium

### Internship (bachelor) at Semicon Sp. z o.o.

6 weeks, Summer 2016   Warsaw, Poland

## SKILLS

### Languages

| | |
|---|---|
| Dutch: native | French: fair |
| English: fluent | German: basic |

### Programming

- Proficient: Python, PyTorch, Numpy, OpenCV, Java, MATLAB, GIT
- Familiar: C/C++, Bash, CUDA, Scikit-learn
- Beginner: Tensorflow, Keras, .NET C#
- Electronics: basic VHDL, PCB design (Altium), Altera FPGA, various microcontrollers
- WebDev: HTML5, PHP, CSS3, Javascript, MySQL, jQuery, Wordpress, Flask, Bootstrap

## EXTRACURRICULAR

Avid participant in tech events such as hackathons and workshops. When I am not programming, working or composing music, I am probably hiking or skiing in the Alps.

- **Hackathons**: building prototype apps and services in 48 hours
  - Finalist and challenge winner of the IBM Challenge at HackZurich 2018 (Europe's largest hackathon): agricultural risk forecasting
  - Public award and challenge winner at HackTheAlps: ski slope detection from aerial imaging using CNNs
- **AFC Participant and Student Consultant** (2015-2017): IT consulting in a multi-disciplinary team of 5. Leadership, presentation & productivity workshop tracks.
- **Full-stack web developer** (2012-...): development and support of a templating framework. PageRank 5. Over 500.000 visitors and 5 million pageviews since 2012. http://metro-webdesign.info/

## REFERENCES

PhD promotor: Prof. Tinne Tuytelaars
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